1.1 Points, Lines, and Planes

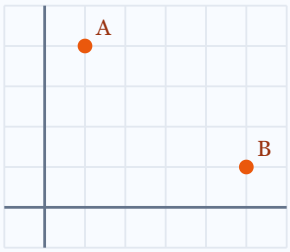
Point, line, and plane are the three undefined terms that every later definition and postulate is built from. Beyond simply naming a ray or a segment, this section pushes into the logic behind collinearity and coplanarity — using postulates to justify why three points determine a plane only when they are noncollinear, and why two planes can only intersect in a line, never in a single stray point. Expect problems that combine several undefined terms in one figure, ask you to name the postulate that justifies an answer, and require reasoning about three-dimensional configurations rather than just flat sketches.

Examples

Example 1: Name a line two ways

Points $A(1, 4)$ and $B(5, 1)$ are graphed. Explain why exactly one line passes through both, then name that line in two ways.

Solution



Exactly one line can be drawn through A and B.

Identify the postulate: the Line Postulate states that through any two distinct points there is exactly one line — once A and B are fixed, the connecting line is uniquely determined.

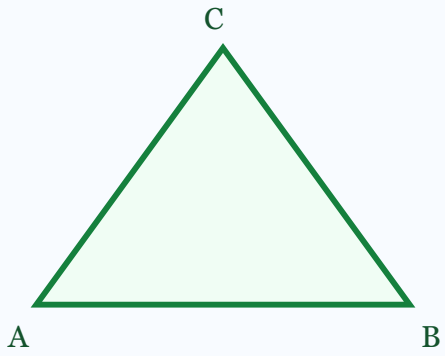
A line extends without end in both directions and has no preferred starting point, so it may be named from either point.

Answer: $\overleftrightarrow{AB}$ or $\overleftrightarrow{BA}$ name the same line.

Example 2: Name a plane and check the condition

Points A, B, C do not lie on one line. Explain why they determine a plane, then name it.

Solution



Three noncollinear points pin down exactly one flat plane.

Check the condition: the Plane Postulate requires the three points to be **noncollinear** — that is given here.

Apply the postulate: three noncollinear points determine exactly one plane containing all of them.

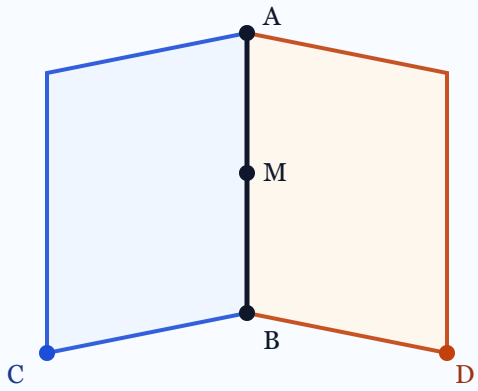
Name the plane using any order of the three letters.

Answer: Plane ABC (equivalently ACB, BAC, etc.).

Example 3: Collinear vs. coplanar across two planes

In the figure, two planes are joined along $\overline{AB}$, with M between A and B, point C in the left plane, and point D in the right plane. (a) Are A, M, B collinear? (b) Are A, B, C, D all coplanar?

Solution



Two planes share the line through A, M, B; C and D sit in different planes.

(a) A, M, B all lie on the shared edge, which is a single line, so **they are collinear**.

(b) C belongs only to the left plane and D belongs only to the right plane; one flat plane cannot contain points from both slanted surfaces at once, so A, B, C, D are **not coplanar**.

Lesson: coplanar must hold for the whole set at once — it is not enough that some triples within it are coplanar.

Example 4: Opposite rays

A, M, B are collinear with M between A and B. Are $\overrightarrow{MA}$ and $\overrightarrow{MB}$ opposite rays? What about $\overrightarrow{AM}$ and $\overrightarrow{MB}$?

Solution



M lies between A and B on one line.

Check the definition: opposite rays must share the same endpoint *and* point in exactly opposite directions along one line.

$\overrightarrow{MA}$ and $\overrightarrow{MB}$ both start at M and head toward opposite ends of the line — both conditions hold, so **they are opposite rays**.

$\overrightarrow{AM}$ starts at A, not M, so it does not share an endpoint with $\overrightarrow{MB}$ — **not opposite rays**, even though all three points lie on the same line.

Answer: $\overrightarrow{MA}, \overrightarrow{MB}$ are opposite; $\overrightarrow{AM}, \overrightarrow{MB}$ are not.

Example 5: Uniqueness of a line

Explain, using a postulate, why two distinct points cannot lie on two different lines.

Solution

State the postulate: the Line Postulate guarantees exactly one line through any two distinct points — the word "exactly" rules out more than one.

Suppose, for contradiction, that two different lines ℓ_1 and ℓ_2 both passed through points A and B. That would contradict the postulate, which promises only one such line.

Conclusion: two distinct points determine one line, and only one — never two.

Example 6: A point off a line determines a plane

Line ℓ contains points D, E, F. Point G is not on ℓ. How many planes contain both ℓ and G? Justify your answer.

Solution

Reduce to the three-point case: G together with any two points of ℓ (say D and E) forms a noncollinear triple, since G is not on ℓ.

Apply the postulate: three noncollinear points determine exactly one plane, and that same plane contains all of ℓ (since $D, E \in \ell$ forces the whole line in, by the Line-in-Plane Postulate).

Answer: Exactly one plane contains both ℓ and G.

Example 7: Two intersecting lines determine a plane

Lines j and k intersect at point P. Explain why j and k must lie in a common plane, and how many such planes exist.

Solution

Pick a second point on each line: $Q \neq P$ on j, and $R \neq P$ on k.

P, Q, R are noncollinear (since Q lies only on j and R lies only on k, and $j \neq k$), so they determine exactly one plane.

That plane contains P and Q, so by the Line-in-Plane Postulate it contains all of line j; it contains P and R, so it contains all of line k.

Conclusion: intersecting lines always lie in exactly one common plane.

Example 8: Counting planes from four points

Four points A, B, C, D are positioned so that no three are collinear, and the four points are not all coplanar. How many different planes are determined by choosing 3 of the 4 points at a time?

Solution

Count the ways to choose 3 points from 4: $\binom{4}{3} = 4$ triples — ABC, ABD, ACD, BCD.

Since no three points are collinear, each triple is noncollinear and determines exactly one plane.

Because the four points are not all coplanar, no single plane can contain more than 3 of them, so the four triples give four *different* planes — like the four triangular faces of a tetrahedron.

Answer: 4 planes.

Example 9: Real-world model — a room corner

In a rectangular room, the floor and two adjacent walls meet at a corner. Model the corner, the wall-to-wall edge, and each flat surface using undefined terms, then describe (a) the intersection of the two walls and (b) the intersection of all three surfaces.

Solution

Model: each flat surface (floor, each wall) is a **plane**; the edge where the two walls meet is a **line**; the corner where all three surfaces meet is a **point**.

(a) Two planes (the two walls) intersect in a line — the vertical edge where the walls meet.

(b) All three planes share only the single corner point, since the vertical wall edge touches the floor at just that one spot.

Takeaway: two planes meet in a line, but three planes in general position meet in a single point.

Example 10: Line–plane intersection

Line t is not contained in plane P, but t does cross through P. What is $t \cap \mathrm{P}$, and why can it not be more than that?

Solution

Since t is not contained in P, at least one point of t lies off the plane. If t shared two or more points with P, the Line-in-Plane Postulate would force *all* of t into P — a contradiction.

So t shares at most one point with P; since we are told it does cross the plane, it shares exactly one.

Answer: $t \cap \mathrm{P}$ is a single point.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Name a line

Name the line through M and N.

Answer: **Verification:**

Problem 2: Name a ray

Write the ray that starts at A and goes through B.

Answer: **Verification:**

Problem 3: Name a segment

Write the segment with endpoints P and Q.

Answer: **Verification:**

Problem 4: Collinear?

Points A, B, C do not lie on one line. Are they collinear?

Answer: **Verification:**

Problem 5: Coplanar?

Three noncollinear points A, B, C. Are they coplanar?

Answer: **Verification:**

Problem 6: Opposite rays

Points R, S, T collinear with S between R and T. Name a pair of opposite rays.

Answer: **Verification:**

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Problem 7: How many planes

How many planes contain three given noncollinear points?

Answer: **Verification:**

Problem 8: How many lines

How many lines contain two given distinct points?

Answer: **Verification:**

Problem 9: Plane intersection

If planes P and Q intersect (and are not equal), what shape is $P \cap Q$?

Answer: **Verification:**

Problem 10: Segment vs ray

Does $\overline{AB}$ have finite length?

Answer: **Verification:**

Problem 11: Ray length

Does $\overrightarrow{AB}$ have finite length?

Answer: **Verification:**

Problem 12: Point off a line

Line ℓ and point $G \notin \ell$ are given. How many planes contain both?

Answer: **Verification:**

Problem 13: Intersecting lines

Lines j and k intersect at P; $Q \in j$ and $R \in k$ with $Q, R \neq P$. Name the plane containing j and k.

Answer: **Verification:**

Problem 14: Challenge — counting planes

Five points are positioned so that no 3 are collinear and no 4 are coplanar. How many planes are determined by choosing 3 points at a time?

Answer: **Verification:**

Problem 15: Challenge — tetrahedron

A tetrahedron has 4 vertices, no 3 collinear and not all 4 coplanar. Using combinations, find the number of edges (lines) and the number of face planes, and explain the pattern.

Answer: **Verification:**

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✓ Answer Key

1. $\overleftrightarrow{MN}$
2. $\overrightarrow{AB}$
3. $\overline{PQ}$
4. No
5. Yes — any 3 points, collinear or not, lie in at least one common plane
6. $\overrightarrow{SR}$ and $\overrightarrow{ST}$
7. Exactly one
8. Exactly one
9. A line
10. Yes
11. No — it extends forever past B
12. Exactly one
13. Plane PQR (or QPR, etc.)
14. $\binom{5}{3} = 10$ planes
15. $\binom{4}{2} = 6$ edges; $\binom{4}{3} = 4$ face planes — edges come from pairs of vertices, faces from triples

Great work — keep practising!